

Public Economics (ECON 131)

Section #9: Public Goods

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Contents

	Page
1 Key Concepts	1
2 Practice Problems	3
2.1 Gruber, Ch.7, Q.13	3
2.2 Gruber, Ch.7, Q.15	5
2.3 Gruber, Ch.7, Q.12 [EXTRA PROBLEM]	6

1 Key Concepts

- A pure public good is a good that satisfies the following two conditions:

- Non-rivalry: The consumption of the good by one individual doesn't affect other's opportunity to consume the good.
- Non-excludability: Individuals cannot be excluded from consuming the good.

- **Samuelson Rule**: Since a public good can simultaneously benefit many individuals, the optimal provision of a public good is achieved when the marginal cost of provision is set equal to the sum of the individuals' marginal rate of substitutions for the good.
- The inability to exclude people from enjoying a public good creates the incentive for individuals to consume it without paying, i.e. **free riding**. This results in the underprovision of public goods in the private market.

Samuelson rule (mathematically)

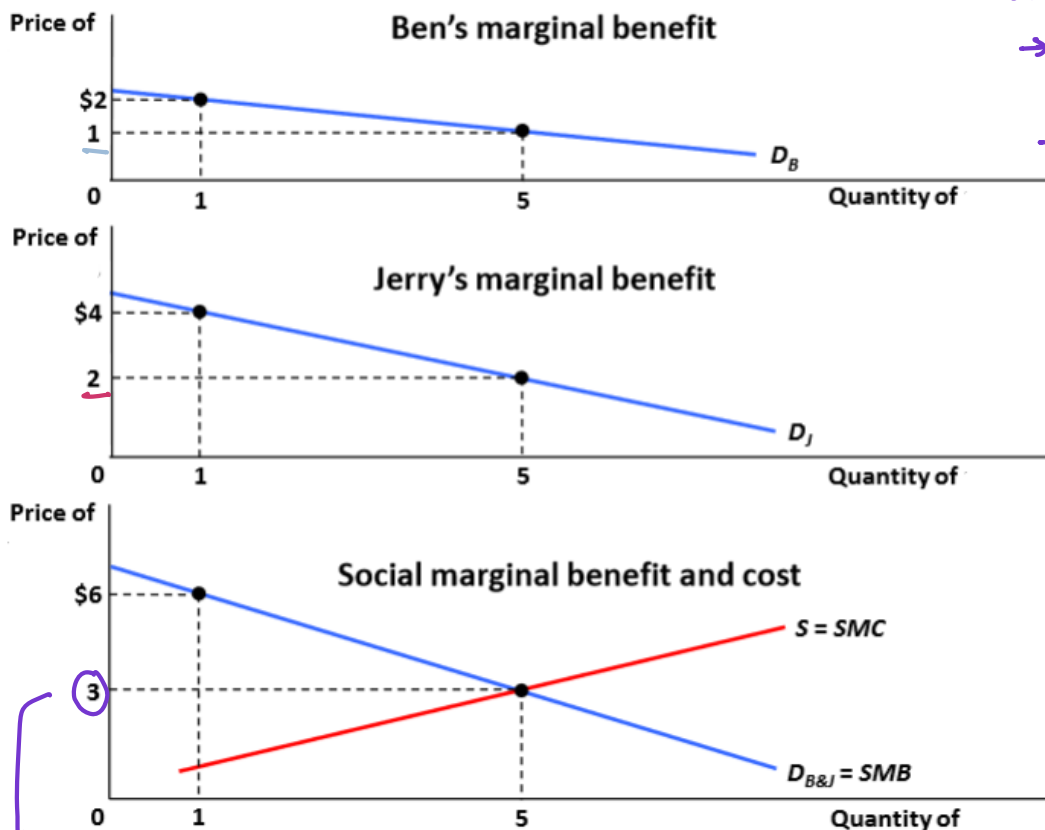
$$\underset{\substack{\downarrow \\ \text{the marginal cost of providing} \\ \text{an additional unit of public} \\ \text{good } m}}{MC_m} = \sum_i \underset{\substack{\downarrow \\ \text{the \# of private good } x \text{ that consumer } i \\ \text{is willing to give up for an additional} \\ \text{unit of public good } m \text{ [varies by } i \text{ + their preferences]}}}{MRS_{m,x}^i}$$

in net, society is willing to give up a $\# (MRS_{m,x}^1 + MRS_{m,x}^2 + \dots)$ of good x_1 for an additional unit of public good m

→ why do we sum vertically? has to do w/ the fact that, unlike private goods, public goods are non-rivalrous

The **Social Demand Curve** is the aggregated demand for a good and can be found by summing the individual demand curves *vertically* for public goods (as opposed to horizontally for private goods).

street lighting as a public good
→ lights road for everyone
→ one more person walking doesn't "use up" the light



$$3 = \underline{1} + \underline{2}$$

this comes from Ben + Jerry's individual demand curves

2 Practice Problems

2.1 Gruber, Ch.7, Q.13

The town of Springfield has two residents: Homer and Bart. The town currently funds its fire department solely from the individual contributions of these residents. Each of the two residents has a utility function over private goods (X_i) and total firefighters (M) of the form

$$U_i = 4 \ln(X_i) + 2 \ln(M)$$

where $i = B, H$. The total provision of firefighters hired, M , is the sum of the number hired by each of the two persons: $M = M_H + M_B$. Homer and Bart both have income of \$100, and the price of both the private good and a firefighter is \$1. Thus, they are limited to providing between 0 and 100 firefighters.

- (a) How many firefighters are hired if the government does not intervene? How many are paid for by Homer? By Bart?

what is Bart's budget constraint? income = consumption $\Rightarrow 100 = 1 \cdot X_B + 1 \cdot M_B$ (both goods worth a dollar)

maximize Bart's utility! $V_B = 4 \ln(X_B) + 2 \ln(M) = 4 \ln(100 - M_B) + 2 \ln(M_H + M_B)$

$$\frac{\partial V_B}{\partial M_B} = 4 \cdot \frac{1}{100 - M_B} \cdot -1 + \frac{2}{M_H + M_B} = 0 \Rightarrow \frac{2}{M_H + M_B} = \frac{4}{100 - M_B} \Rightarrow 2(100 - M_B) = 4(M_H + M_B)$$

$$\Rightarrow 100 - M_B = 2M_H + 2M_B \Rightarrow 100 - 2M_H = 3M_B \Rightarrow M_B = \frac{100 - 2M_H}{3}$$

what is Homer's BC? $100 = X_H + M_H$

utility? $V_H = 4 \ln(100 - M_H) + 2 \ln(M_H + M_B)$

> we have symmetry!

$$M_H = \frac{100 - 2M_B}{3}$$

finally, plug in M_H into M_B , such that

$$M_B = \frac{100 - 2 \left[\frac{100 - 2M_B}{3} \right]}{3} \Rightarrow \left[3M_B = 100 - \frac{200}{3} + \frac{4M_B}{3} \right] \cdot 3$$

$$\Rightarrow 9M_B = 300 - 200 + 4M_B$$

$$\Rightarrow 5M_B = 100$$

$$\Rightarrow M_B^* = 20$$

$$M_H^* = \frac{100 - 2(20)}{3} = 20$$

$$M^* = M_B^* + M_H^* = 20 + 20 = 40$$

if gov't does not intervene, 40 firefighters will be hired. 20 by Homer and 20 by Bart

- (b) What is the socially optimal number of firefighters? ^{yes!} If your answer differs from part (a), why? Would government intervention increase Bart's utility?

substitution

social planner maximizes an aggregate of utilities, denoted by SU

$$SU = U_B + U_H = 4 \ln(100 - M_B) + 2 \ln(M_H + M_B) + 4 \ln(100 - M_H) + 2 \ln(M_H + M_B)$$

$$\frac{\partial u}{\partial M_B} = \frac{4}{100 - M_B} (-1) + \frac{2}{M_H + M_B} + \frac{2}{M_H + M_B} = 0 \Rightarrow \frac{4}{100 - M_B} = \frac{4}{M_H + M_B}$$

$$\Rightarrow 4(M_H + M_B) = 4(100 - M_B) \Rightarrow 2M_B = 100 - M_H \Rightarrow M_B = \frac{100 - M_H}{2}$$

$$\frac{\partial u}{\partial M_H} \stackrel{(\text{the same})}{=} \frac{4}{100 - M_H} (-1) + \frac{2}{M_H + M_B} + \frac{2}{M_H + M_B} = 0 \Rightarrow \frac{4}{100 - M_H} = \frac{4}{M_H + M_B}$$

$$\Rightarrow 4(M_H + M_B) = 4(100 - M_H) \Rightarrow 2M_H = 100 - M_B \Rightarrow M_H = \frac{100 - M_B}{2}$$

plug M_H into M_B , we get that

$$M_B = \frac{100 - \left[\frac{100 - M_B}{2} \right]}{2} \Rightarrow \left[2M_B = 100 - \left(\frac{100 - M_B}{2} \right) \right] \cdot 2$$

$$\Rightarrow 4M_B = 200 - 100 + M_B \Rightarrow 3M_B = 100 \Rightarrow M_B = \frac{100}{3} = 33.33$$

$$\stackrel{\text{the same}}{M_H} = 33.33$$

$$M_{SO} = M_B + M_H = 33.33 + 33.33 = 66.67$$

use Samuelson rule!

(production trade-off, to produce more M , give up resources to produce good X)

$$U_i = 4 \ln(X_i) + 2 \ln(M)$$

$$\underline{MRS_{M,X}^B} + \underline{MRS_{M,X}^H} = \sum MRS_{M,X}^i = MC_M = \frac{P_M}{P_X} = 1$$

$$\underline{MRS_{M,X}^B} = \frac{MU_M}{MU_X} = \frac{\frac{2}{M}}{\frac{4}{X_B}} = \frac{\frac{2}{M}}{\frac{4}{100 - M_B}}$$

$$\underline{MRS_{M,X}^H} = \frac{\frac{2}{M}}{\frac{4}{100 - M_H}} \quad (\text{symmetry})$$

$$1 = \frac{\frac{2}{M}}{\frac{4}{100 - M_B}} + \frac{\frac{2}{M}}{\frac{4}{100 - M_H}} = \frac{2}{M} \cdot \frac{100 - M_B}{4} + \frac{2}{M} \cdot \frac{100 - M_H}{4} = \frac{100 - M_B + 100 - M_H}{2M}$$

$$\Rightarrow 2M = 200 - (M_B + M_H) = 200 - M \Rightarrow 3M = 200$$

$$M^* = \frac{200}{3} = 66.67$$

2.2 Gruber, Ch.7, Q.15

Consider an economy with three types of individuals, differing only with respect to their preferences for monuments. Individuals of the first type get a fixed benefit of 100 from the mere existence of monuments, whatever their number. Individuals of the second and third type get benefits according to $B_{II} = 200 + 30M - 1.5M^2$ and $B_{III} = 150 + 90M - 4.5M^2$, where M denotes the number of monuments in the city. Assume that there are 50 people of each type. Monuments cost \$3,600 each to build. How many monuments should be built?

$$SMC = SMB$$

what is the SMC? $SMC = MC = 3600$

we want to find SMB, but we have three types of individuals w/ different benefits they derive from monuments.

$$\textcircled{I} \quad B_I = 100$$

$$MB_I = 0$$

$$\textcircled{II} \quad B_{II} = 200 + 30M - 1.5M^2$$

$$MB_{II} = 30 - 3M$$

$$\textcircled{III} \quad B_{III} = 150 + 90M - 4.5M^2$$

$$MB_{III} = 90 - 9M$$

$$SMC = SMB$$

$$3600 = \cancel{50(0)} + 50(30 - 3M) + 50(90 - 9M)$$

$$3600 = 50[30 - 3M + 90 - 9M]$$

$$3600 = 50[120 - 12M]$$

$$72 = 120 - 12M$$

$$12M = 48$$

$$M = 4$$

2.3 Gruber, Ch.7, Q.12 [EXTRA PROBLEM]

Andrew, Beth, and Cathy live in Lindhville. Andrew's demand for bike paths, a public good, is given by $Q = 12 - 2P$. Beth's demand is $Q = 18 - P$, and Cathy's is $Q = 8 - P/3$. The marginal cost of building a bike path is $MC = 21$. The town government decides to use the following procedure for deciding how many paths to build. It asks each resident how many paths they want, and it builds the largest number asked for by any resident. To pay for these paths, it then taxes Andrew, Beth, and Cathy the prices a , b , and c per path, respectively, where $a + b + c = MC$. (The residents know these tax rates before stating how many paths they want.)

- (a) If the taxes are set so that each resident shares the cost evenly ($a = b = c$), how many paths will get built?

if $a = b = c$, then $a + b + c = 3a = MC \Rightarrow 3a = 21$ and $a = b = c = 7$

first, marginal benefits (inverse demand)

$$2P_a = 12 - Q_a$$

$$P_a = \frac{12 - Q_a}{2}$$

$$7 = 6 - \frac{1}{2}Q_a$$

$$1 = -\frac{1}{2}Q_a$$

$Q_a = -2$
so no bike paths for Andrew

$$P_b = 18 - Q_b$$

$$7 = 18 - Q_b$$

$$\Rightarrow Q_b = 11$$

Beth wants 11 bike paths

11 will get built

$$\frac{1}{3}P_c = 8 - Q_c$$

$$P_c = 24 - 3Q_c$$

$$7 = 24 - 3Q_c$$

$$3Q_c = 17$$

$$Q_c = \frac{17}{3} = 5\frac{2}{3} = 5.67$$

Cathy wants 5.67 bike paths

- (b) Show that the government can achieve the social optimum by setting the correct tax prices a , b , and c . What prices should it set?

$$SMC = SMB$$

$$21 = MB_a + MB_b + MB_c = 6 - \frac{1}{2}Q + 18 - Q + 24 - 3Q$$

$$\Rightarrow 21 = (24 + 18 + 6) - (\frac{1}{2}Q + \frac{2}{2}Q + \frac{6}{2}Q)$$

$$\Rightarrow 21 = 48 - \frac{9}{2}Q$$

$$\Rightarrow \frac{9}{2}Q = 27$$

$$\Rightarrow Q = 6$$

recall here that MB is also the inverse demand curve (or isolate P)

that said, we want to tax prices so that nobody will want more than 6 units (and someone will want exactly 6 units)

builds the largest number asked.

$$a = 6 - \frac{1}{2}(6)$$

$$a = 6 - 3$$

$$a = 3$$

$$b = 18 - 6$$

$$b = 12$$

$$c = 24 - 3(6)$$

$$c = 24 - 18$$

$$c = 6$$

$3 + 12 + 6 = 21 = MC$ these tax rates are just enough to cover MC